

EJERCICIOS DE REDES DOCKER

✓ Ejercicio 1 — Crear tres redes Docker personalizadas

Crea tres redes Docker, cada una con su propia IP y rango.

● Red A — redA Subred: 10.10.10.0/24 Gateway: 10.10.10.1 Rango: 10.10.10.50 – 10.10.10.60 ● Red B — redB Subred: 20.20.20.0/24 Gateway: 20.20.20.1 Rango: 20.20.20.100 – 20.20.20.120 ● Red C — redC Subred: 30.30.30.0/24 Gateway: 30.30.30.1 Rango: 30.30.30.150 – 30.30.30.160

Creacion de redes.

```
Administrador: Símbolo del sistema
Microsoft Windows [Versión 10.0.26200.6899]
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C:\Windows\System32>docker network create --subnet=10.10.10.0/24 --ip-range=10.10.10.48/28 --gateway=10.10.10.1 redA
86660e5bee693f568e8287bf5d0d630553f4c50d6d601d6084508aa897d19266

C:\Windows\System32>docker network create --subnet=20.20.20.0/24 --ip-range=20.20.20.96/27 --gateway=20.20.20.1 redB
55eca0e8aa8b2f29781e984cf21d9742f1c86e7340957ec6ab97f4ccb2a7375e

C:\Windows\System32>docker network create --subnet=30.30.30.0/24 --ip-range=30.30.30.128/26 --gateway=30.30.30.1 redC
76e02cef39f322a5120b52d4e597e75b41f37bf0f61dc9cf960254dc845f44c0
```

Red A

```
C:\Windows\System32>docker network inspect redA
[
  {
    "Name": "redA",
    "Id": "86660e5bee693f568e8287bf5d0d630553f4c50d6d601d6084508aa897d19266",
    "Created": "2025-11-11T18:42:19.179067933Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "10.10.10.0/24",
          "IPRange": "10.10.10.48/28",
          "Gateway": "10.10.10.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {}
  }
]
```

Red B

```
C:\Windows\System32>docker network inspect redB
[
  {
    "Name": "redB",
    "Id": "55eca0e8aa8b2f29781e984cf21d9742f1c86e7340957ec6ab97f4ccb2a7375e",
    "Created": "2025-11-11T18:42:52.289963241Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "20.20.20.0/24",
          "IPRange": "20.20.20.96/27",
          "Gateway": "20.20.20.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {}
  }
]
```

Red C

```
C:\Windows\System32>docker network inspect redC
[
  {
    "Name": "redC",
    "Id": "76e02cef39f322a5120b52d4e597e75b41f37bf0f61dcbcf960254dc845f44c0",
    "Created": "2025-11-11T18:42:56.478800639Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "30.30.30.0/24",
          "IPRange": "30.30.30.128/26",
          "Gateway": "30.30.30.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {}
  }
]
```

- ☒ Ejercicio 2 — Lanzar DOS contenedores en cada red (todos distintos) Debes usar SEIS contenedores, todos con un sistema operativo diferente.
- ☒ En redA deben lanzarse: Contenedor Imagen SO a1 ubuntu Ubuntu a2 fedora Fedora
- ☒ En redB deben lanzarse: Contenedor Imagen SO b1 debian Debian b2 centos:7 CentOS
- ☒ En redC deben lanzarse: Contenedor Imagen SO c1 alpine Alpine c2 rockylinux Rocky Linux

Administrador: Símbolo del sistema

```
C:\Windows\System32>docker run -d --name a1 --network redA ubuntu sleep infinity
94aad1737a99fe4769a938d7f2265e0b1cd4e6b5614236574bb9c4380d3f08eb

C:\Windows\System32>docker run -d --name a2 --network redA fedora sleep infinity
b42c9512f8d238b8789a64e1c80c267850b7ce478301f8e4b14e6e4201daeb53

C:\Windows\System32>docker run -d --name b1 --network redB debian sleep infinity
6eeea11fec6c6e52e606b83c2021903df340a277baf2c238e3a0f1e431342e8c

C:\Windows\System32>docker run -d --name b2 --network redB centos:7 sleep infinity
a7c495910b018a95a028dd2ee32d1eca6a1c013cc4229328efb6d07b557b65e4

C:\Windows\System32>docker run -d --name c1 --network redC alpine sleep infinity
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
2d35ebdb57d9: Pull complete
Digest: sha256:4b7ce07002c69e8f3d704a9c5d6fd3053be500b7f1c69fc0d80990c2ad8dd412
Status: Downloaded newer image for alpine:latest
955b9603a1e3297e555337d529516ece9a155197a6d0e201697b59d15f3ffe12
```

Administrador: Símbolo del sistema

```
C:\Windows\System32>docker run -d --name c2 --network redC rockylinux/rockylinux sleep infinity
Unable to find image 'rockylinux/rockylinux:latest' locally
latest: Pulling from rockylinux/rockylinux
71cc2ddb2ecf: Pull complete
Digest: sha256:fc370d748f4cd1e6ac3d1b6460fb82201897fa15a16f43e947940df5aca1a56e
Status: Downloaded newer image for rockylinux/rockylinux:latest
89a5098ef137babcdca56d57ebfbecb57af5916196a87eb21eab3a901ddbc6ca

C:\Windows\System32>
```

☒ Ejercicio 3 — Crear un contenedor conectado a las TRES redes Crea un contenedor llamado multired3 basado en Fedora que debe pertenecer a: redA redB redC Tareas: Lanzarlo inicialmente en redA. Conectarlo después a redB y redC. Verificar 3 interfaces con ip addr.

```

Administrador: Símbolo del sistema
Microsoft Windows [Versión 10.0.26200.6899]
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C:\Windows\System32>docker run -d --name multired3 --network redA fedora sleep infinity
1bd9626ed5476840edd8136bcfb7782768963d8c460f13090b451fd87297442c

C:\Windows\System32>docker network connect redB multired3

C:\Windows\System32>docker network connect redC multired3

C:\Windows\System32>

```

Instalamos los paquetes de red

```

Administrador: Símbolo del sistema - docker exec -it multired3 bash -lc "yum update && yum install -y iputils-ping dnsutils curl"

C:\Windows\System32>docker exec -it multired3 bash -lc "yum update && yum install -y iputils-ping dnsutils curl"
Updating and loading repositories:
Fedora 42 openh264 (From Cisco) - x86_64                               100% | 2.4 KiB/s | 5.8 KiB | 00m02s
Fedora 42 - x86_64 - Updates                                         15% [===] | 152.2 KiB/s | 1.1 MiB | 00m41s
Fedora 42 - x86_64                                                  5% [=] | 92.2 KiB/s | 2.1 MiB | 06m37s

```

Ip address

```

Administrador: Símbolo del sistema

C:\Windows\System32>docker exec multired3 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: eth0@if14: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether be:df:a4:e0:77:04 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.10.10.50/24 brd 10.10.10.255 scope global eth0
        valid_lft forever preferred_lft forever
3: eth1@if15: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether d2:86:1f:ff:9e:ee brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 20.20.20.98/24 brd 20.20.20.255 scope global eth1
        valid_lft forever preferred_lft forever
4: eth2@if16: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 4a:60:30:83:38:ec brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 30.30.30.130/24 brd 30.30.30.255 scope global eth2
        valid_lft forever preferred_lft forever

```

- ☒ Ejercicio 4 — Crear un contenedor conectado a DOS redes Crea un contenedor llamado multired2 basado en distroless o busybox que debe pertenecer a: redA redC Tareas: Implantación de Sistemas Operativos 1º ASIR Profesora: Ana Fuentes 2/3 Inicialmente lanzarlo en redC. Añadirlo a redA. Verificar interfaces: 2 direcciones IP

```
Administrador: Símbolo del sistema

C:\Windows\System32>docker run -d --name multired2 --network redC busybox sleep infinity
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
59838ecfec5: Pull complete
Digest: sha256:e3652a00a2fabd16ce889f0aa32c38eec347b997e73bd09e69c962ec7f8732ee
Status: Downloaded newer image for busybox:latest
bb9d8e6edd5c99abc385256906b7677adb04866f2c755ad525ca39329b1b032f

C:\Windows\System32>docker network connect redA multired2

C:\Windows\System32>
```

```
C:\Windows\System32>docker exec multired2 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0@if17: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1500 qdisc noqueue
    link/ether 0a:89:9d:d7:0c:fd brd ff:ff:ff:ff:ff:ff
    inet 30.30.30.131/24 brd 30.30.30.255 scope global eth0
        valid_lft forever preferred_lft forever
3: eth1@if18: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1500 qdisc noqueue
    link/ether 4a:e7:fb:e0:2a:5e brd ff:ff:ff:ff:ff:ff
    inet 10.10.10.51/24 brd 10.10.10.255 scope global eth1
        valid_lft forever preferred_lft forever
```

- ☒ Ejercicio 5 — Comprobación de conectividad ☒ 5.1 Comunicaciones dentro de la misma red a1 ↔ a2 b1 ↔ b2 c1 ↔ c2

```
Administrador: Símbolo del sistema

C:\Windows\System32>docker exec a1 ping -c 3 a2
PING a2 (10.10.10.49) 56(84) bytes of data.
64 bytes from a2.redA (10.10.10.49): icmp_seq=1 ttl=64 time=0.060 ms
64 bytes from a2.redA (10.10.10.49): icmp_seq=2 ttl=64 time=0.055 ms
64 bytes from a2.redA (10.10.10.49): icmp_seq=3 ttl=64 time=0.059 ms

--- a2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2043ms
rtt min/avg/max/mdev = 0.055/0.058/0.060/0.002 ms

C:\Windows\System32>
C:\Windows\System32>docker exec b1 ping -c 3 b2
OCI runtime exec failed: exec failed: unable to start container process: exec: "ping": executable file not found in $PATH: unknown

C:\Windows\System32>docker exec b2 ping -c 3 b1
PING b1 (20.20.20.96) 56(84) bytes of data.
64 bytes from b1.redB (20.20.20.96): icmp_seq=1 ttl=64 time=0.164 ms
64 bytes from b1.redB (20.20.20.96): icmp_seq=2 ttl=64 time=0.055 ms
64 bytes from b1.redB (20.20.20.96): icmp_seq=3 ttl=64 time=0.047 ms

--- b1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2043ms
rtt min/avg/max/mdev = 0.047/0.088/0.164/0.054 ms

C:\Windows\System32>
C:\Windows\System32>

C:\Windows\System32>docker exec c1 ping -c 3 c2
PING c2 (30.30.30.129) 56(84) bytes of data.
64 bytes from c2.redC (30.30.30.129): icmp_seq=1 ttl=64 time=0.136 ms
64 bytes from c2.redC (30.30.30.129): icmp_seq=2 ttl=64 time=0.061 ms
64 bytes from c2.redC (30.30.30.129): icmp_seq=3 ttl=64 time=0.054 ms

--- c2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2028ms
rtt min/avg/max/mdev = 0.054/0.083/0.136/0.037 ms
```

- ☒ 5.2 Comunicaciones de los contenedores multired ☒ multired3 (opensuse) debe alcanzar: a1 (Ubuntu), a2 (Fedora)

Administrador: Símbolo del sistema

```
C:\Windows\System32>docker exec multired3 ping -c 3 a1
PING a1 (10.10.10.48) 56(84) bytes of data.
64 bytes from a1.redA (10.10.10.48): icmp_seq=1 ttl=64 time=0.178 ms
64 bytes from a1.redA (10.10.10.48): icmp_seq=2 ttl=64 time=0.086 ms
64 bytes from a1.redA (10.10.10.48): icmp_seq=3 ttl=64 time=0.050 ms

--- a1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2012ms
rtt min/avg/max/mdev = 0.050/0.104/0.178/0.053 ms

C:\Windows\System32>docker exec multired3 ping -c 3 a2
PING a2 (10.10.10.49) 56(84) bytes of data.
64 bytes from a2.redA (10.10.10.49): icmp_seq=1 ttl=64 time=0.108 ms
64 bytes from a2.redA (10.10.10.49): icmp_seq=2 ttl=64 time=0.052 ms
64 bytes from a2.redA (10.10.10.49): icmp_seq=3 ttl=64 time=0.031 ms

--- a2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2053ms
rtt min/avg/max/mdev = 0.031/0.063/0.108/0.032 ms

C:\Windows\System32>
```

b1 (Debian), b2 (CentOS)

Administrador: Símbolo del sistema

```
C:\Windows\System32>docker exec multired3 ping -c 3 b1
PING b1 (20.20.20.96) 56(84) bytes of data.
64 bytes from b1.redB (20.20.20.96): icmp_seq=1 ttl=64 time=0.155 ms
64 bytes from b1.redB (20.20.20.96): icmp_seq=2 ttl=64 time=0.084 ms
64 bytes from b1.redB (20.20.20.96): icmp_seq=3 ttl=64 time=0.046 ms

--- b1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.046/0.095/0.155/0.045 ms

C:\Windows\System32>docker exec multired3 ping -c 3 b2
PING b2 (20.20.20.97) 56(84) bytes of data.
64 bytes from b2.redB (20.20.20.97): icmp_seq=1 ttl=64 time=0.228 ms
64 bytes from b2.redB (20.20.20.97): icmp_seq=2 ttl=64 time=0.056 ms
64 bytes from b2.redB (20.20.20.97): icmp_seq=3 ttl=64 time=0.050 ms

--- b2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2049ms
rtt min/avg/max/mdev = 0.050/0.111/0.228/0.082 ms

C:\Windows\System32>
```

c1 (Alpine), c2 (Rocky)

```

C:\Windows\System32>docker exec multired3 ping -c 3 c1
PING c1 (30.30.30.128) 56(84) bytes of data.
64 bytes from c1.redC (30.30.30.128): icmp_seq=1 ttl=64 time=0.205 ms
64 bytes from c1.redC (30.30.30.128): icmp_seq=2 ttl=64 time=0.048 ms
64 bytes from c1.redC (30.30.30.128): icmp_seq=3 ttl=64 time=0.080 ms

--- c1 ping statistics ---
8 packets transmitted, 3 received, 0% packet loss, time 2034ms
rtt min/avg/max/mdev = 0.048/0.111/0.205/0.067 ms

C:\Windows\System32>docker exec multired3 ping -c 3 c2
PING c2 (30.30.30.129) 56(84) bytes of data.
64 bytes from c2.redC (30.30.30.129): icmp_seq=1 ttl=64 time=0.088 ms
64 bytes from c2.redC (30.30.30.129): icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from c2.redC (30.30.30.129): icmp_seq=3 ttl=64 time=0.072 ms

--- c2 ping statistics ---
8 packets transmitted, 3 received, 0% packet loss, time 2029ms
rtt min/avg/max/mdev = 0.049/0.069/0.088/0.016 ms

C:\Windows\System32>

```

 multired2 (busybox/distroless) debe alcanzar solo: a1, a2 (por redA)

```

C:\Windows\System32>docker exec multired2 ping -c 3 a1
PING a1 (10.10.10.48): 56 data bytes
64 bytes from 10.10.10.48: seq=0 ttl=64 time=0.144 ms
64 bytes from 10.10.10.48: seq=1 ttl=64 time=0.063 ms
64 bytes from 10.10.10.48: seq=2 ttl=64 time=0.063 ms

--- a1 ping statistics ---
8 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.063/0.090/0.144 ms

C:\Windows\System32>docker exec multired2 ping -c 3 a2
PING a2 (10.10.10.49): 56 data bytes
64 bytes from 10.10.10.49: seq=0 ttl=64 time=0.104 ms
64 bytes from 10.10.10.49: seq=1 ttl=64 time=0.079 ms
64 bytes from 10.10.10.49: seq=2 ttl=64 time=0.062 ms

--- a2 ping statistics ---
8 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.062/0.081/0.104 ms

C:\Windows\System32>

```

c1, c2 (por

```
Administrador: Símbolo del sistema

C:\Windows\System32>docker exec multired2 ping -c 3 c1
PING c1 (30.30.30.128): 56 data bytes
64 bytes from 30.30.30.128: seq=0 ttl=64 time=0.121 ms
64 bytes from 30.30.30.128: seq=1 ttl=64 time=0.065 ms
64 bytes from 30.30.30.128: seq=2 ttl=64 time=0.068 ms

--- c1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.065/0.084/0.121 ms

C:\Windows\System32>docker exec multired2 ping -c 3 c2
PING c2 (30.30.30.129): 56 data bytes
64 bytes from 30.30.30.129: seq=0 ttl=64 time=0.106 ms
64 bytes from 30.30.30.129: seq=1 ttl=64 time=0.064 ms
64 bytes from 30.30.30.129: seq=2 ttl=64 time=0.070 ms

--- c2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.064/0.080/0.106 ms

C:\Windows\System32>
```

redC)

- ☒ 5.3 Aislamiento esperado a1 NO debe alcanzar b1 b2 NO debe alcanzar c1 a2 NO debe alcanzar c2

```
Administrador: Símbolo del sistema

C:\Windows\System32>docker exec a1 ping -c 3 b1
ping: b1: No address associated with hostname

C:\Windows\System32>docker exec b2 ping -c 3 c1
ping: c1: Name or service not known

C:\Windows\System32>docker exec a2 ping -c 3 c2
ping: c2: No address associated with hostname

C:\Windows\System32>
```